

03 — IVR Nodeflow Design (Call Channel)

Audience: ECC Admin, Exotel Professional Services **Purpose:** Route inbound support calls to tier-appropriate queues based on live Salesforce lookup of caller's Account tier. **Platform:** Exotel 4x or 6x (Harmony) — documented capabilities only.

1. Scope & Capability Statement

Documented Capability

Per docs.exotel.com/contact-center and docs.exotel.com/harmony:

- **IVR nodeflow (callflow)** — supported on both 4x and 6x
- **HTTP action / Webhook node** in IVR — supported for pulling data mid-call from external systems
- **Variable capture and branching** in IVR — standard
- **Queue assignment / skill-based routing** — standard

Not Documented / Requires PS Confirmation

- Exact request timeout behavior and retry semantics of the HTTP action node
- Maximum payload size for variables passed to agent via CTI call context
- Exact syntax for OAuth 2.0 client credentials handshake embedded in IVR (most customers use middleware for this)

For anything marked "requires PS confirmation," see [05 — PS Engagement Brief](#).

2. Architecture Options

Option A — Direct IVR → Salesforce (Simpler, Riskier)

```
Caller → Exotel IVR → HTTP GET to Salesforce /services/data/v59.0/query/
                        ↓
                    (OAuth token must be pre-negotiated and injected as header)
                        ↓
                Response parsed in IVR
                        ↓
                Queue assignment
```

Pros: Fewer moving parts, no middleware to maintain. **Cons:**

- OAuth token refresh inside IVR is non-trivial (tokens expire every 2 hours)
- Salesforce API errors/timeouts hit caller wait time directly
- Hard to add caching
- Security: OAuth credentials stored in IVR config = audit risk

Option B — IVR → Middleware → Salesforce (Recommended)

```
Caller → Exotel IVR → HTTP GET to middleware endpoint (sub-100ms cached response)
                        ↓
                Token mgmt, caching, retries
```

↓
Salesforce /services/data/v59.0/query/

Pros:

- Token refresh handled outside IVR
- 5-minute in-memory cache on phone → tier mapping dramatically reduces SF API load
- Single place to add retries, circuit breakers
- Security: OAuth creds live in middleware vault, not IVR config
- Easier to add business logic (override lists, test numbers)

Cons: Extra service to deploy and monitor.

Recommendation: Option B

Middleware can be minimal — a 200-line Node.js / Python service with:

- /lookup?phone=XXX endpoint
- In-memory LRU cache (5-min TTL)
- Salesforce OAuth client credentials flow with refresh
- Deploy on AWS Lambda + API Gateway, or existing internal service infra

This doc specifies Option B design. Option A is mentioned only as fallback if middleware is not approved.

3. Middleware Specification

3.1 — API Contract

Endpoint:

```
GET https://support-router.internal.exotel.com/api/v1/account-tier?phone={e164_phone}
Headers:
  Authorization: Bearer <static-ivr-shared-secret>
  X-Request-Id: <ivr-session-id>
```

Success Response (200):

```
{
  "matched": true,
  "account_id": "001XXXXXXXXXXXXX",
  "account_name": "Acme Corp",
  "support_tier": "STRATEGIC",
  "rag_status": "GREEN",
  "csm_name": "Priya Sharma",
  "csm_email": "priya@exotel.com",
  "support_group": "Normal Support",
  "is_hypercure": false,
  "lookup_source": "cache" | "salesforce",
  "lookup_ms": 42
}
```

No Match (200):

```
{
  "matched": false,
  "phone_queried": "+919876543210"
}
```

Error (500 / timeout):

```
{
  "error": "salesforce_timeout",
  "message": "Salesforce API did not respond within 2s",
  "fallback_tier": "STANDARD"
}
```

IVR should treat **any non-200 response** as `tier = STANDARD` and route accordingly.

3.2 — Salesforce Query Used by Middleware

```
SELECT
  Id, Name,
  Support_Tier__c,
  RAG__c,
  Hypercare_active__c,
  Support_Group__c,
  Account_CSM__r.Name,
  Account_CSM__r.Email
FROM Account
WHERE (Phone = :normalizedPhone OR Phone = :plus91Phone OR Phone = :bareIndianPhone)
  AND IsDeleted = FALSE
LIMIT 1
```

Middleware normalizes phone into 3 formats (`+919876543210` , `919876543210` , `9876543210`) and passes all three as bind variables.

Fallback query (if Account.Phone miss): check Contact.Phone

```
SELECT AccountId, Account.Support_Tier__c, Account.RAG__c, Account.Name,
  Account.Account_CSM__r.Name, Account.Account_CSM__r.Email
FROM Contact
WHERE (Phone = :normalizedPhone OR MobilePhone = :normalizedPhone OR
  Phone = :plus91Phone OR MobilePhone = :plus91Phone)
  AND AccountId != NULL
LIMIT 1
```

3.3 — Middleware Authentication to Salesforce

Use **OAuth 2.0 Client Credentials Flow** via a Salesforce Connected App.

Connected App Configuration (SF Admin task):

Setting	Value
Name	ECC Revenue Router
API Enabled	Yes
OAuth Scopes	api, refresh_token, offline_access
Require Secret for Web Server Flow	Yes
Client Credentials Flow Run-As User	A dedicated integration user with read-only access to Account, Contact
IP Restrictions	Middleware service IP range

Token Request (middleware):

```
POST https://ameyo.my.salesforce.com/services/oauth2/token
Body: grant_type=client_credentials
      &client_id=<consumer_key>
      &client_secret=<consumer_secret>
```

Token valid for 2 hours. Middleware refreshes on 401 response. Keep a 5-min buffer before expiry to avoid race.

3.4 — Middleware Caching

- **Cache key:** normalized phone
- **Cache value:** full response payload
- **TTL:** 300 seconds (5 min)
- **Invalidation:** none (naturally expires)

If Account tier changes mid-call-cycle, propagation takes up to 5 minutes. Acceptable given tier changes are infrequent.

4. IVR Nodeflow Design

4.1 — Node Map

```
[START: Inbound Call]
  |
  ▼
[Node 1] CAPTURE CALLER ID (ANI)
  {{caller_number}} auto-captured
  |
  ▼
[Node 2] NORMALIZE (optional regex strip)
  strip '+', '91' prefix, spaces – store as {{normalized_phone}}
  |
  ▼
[Node 3] HTTP GET to middleware
  URL: https://support-router.internal.exotel.com/api/v1/account-tier?phone=
```

```

{{normalized_phone}}
Headers: Authorization: Bearer <shared-secret>
Timeout: 2s
On timeout/error → Node 5e (Standard queue)
|
▼
[Node 4] PARSE RESPONSE → extract support_tier, account_name, csm_name, rag_status
|
▼
[Node 5] BRANCH on {{support_tier}}
├ "HYPERCARE" → Node 6a (Hypercare Queue)
├ "STRATEGIC" → Node 6b (Strategic Queue)
├ "PREMIUM" → Node 6c (Premium Queue)
├ "STANDARD" → Node 6d (Standard FLR Queue)
└ "" / null → Node 6e (Unknown/Helpdesk Queue + log)
|
▼
[Node 6x] ANNOUNCE + HOLD
Play tier-specific hold message while waiting
|
▼
[Node 7] TRANSFER TO QUEUE
Pass call context payload to agent
|
▼
[END: Agent answers → CTI screen pop fires]

```

4.2 — Node Detail

Node 1 — Capture Caller ID

- **Node type:** Built-in / automatic
- **Variable:** `{{caller_number}}` — populated by ECC from SIP headers / PSTN signaling
- **Edge cases:**
 - If private / masked CLI → `{{caller_number}}` is blank or `anonymous`
 - Branch: if blank → skip to Node 6e (Unknown queue)

Node 2 — Normalize Phone

- **Node type:** Variable manipulation / set-variable
- **Action:** strip `+`, leading country code `91`, whitespace
- **Logic (pseudo):**

```

{{normalized_phone}} = replace({{caller_number}}, /\s+\-()/g, '')
if length({{normalized_phone}}) == 12 && starts_with({{normalized_phone}},
'91')
    {{normalized_phone}} = substring({{normalized_phone}}, 2)

```

- **Result:** 10-digit Indian mobile format, or original for international

PS confirmation: IVR nodeflow variable manipulation syntax varies between 4x and 6x. PS to provide the exact syntax for the customer's deployment.

Node 3 — HTTP GET to Middleware

- **Node type:** HTTP action / Webhook
- **Method:** GET
- **URL:** `https://support-router.internal.exotel.com/api/v1/account-tier?phone={{normalized_phone}}`
- **Headers:**

`Authorization: Bearer <static-shared-secret-configured-in-IVR>`
`X-Request-Id: {{call_sid}}`
`Accept: application/json`
- **Timeout:** 2 seconds
- **Response parse:** store full response as `{{sf_response}}` (JSON object)
- **On success (2xx):** proceed to Node 4
- **On timeout / 5xx:** proceed directly to Node 6e with `{{support_tier}} = "STANDARD"` and log error

Node 4 — Parse Response

Extract fields into IVR variables:

```
{{support_tier}} = {{sf_response.support_tier}}
{{account_name}} = {{sf_response.account_name}}
{{csm_name}}      = {{sf_response.csm_name}}
{{rag_status}}    = {{sf_response.rag_status}}
{{matched}}       = {{sf_response.matched}}
```

If `{{matched}} = false` → set `{{support_tier}} = "UNKNOWN"` and proceed.

Node 5 — Branch Decision

Conditional routing based on `{{support_tier}}` :

Condition	Branch
<code>{{support_tier}} == "HYPERCARE"</code>	Node 6a
<code>{{support_tier}} == "STRATEGIC"</code>	Node 6b
<code>{{support_tier}} == "PREMIUM"</code>	Node 6c
<code>{{support_tier}} == "STANDARD"</code>	Node 6d
else	Node 6e

Nodes 6a–6e — Queue Assignment

Branch	Queue	Hold Music / IVR Prompt	Expected Wait (target)
6a HYPERCARE	Hypercare Support Queue	"Connecting you to your dedicated support team..."	< 30s

6b STRATEGIC	Strategic Support Queue	"Thank you for calling, connecting to priority support..."	< 60s
6c PREMIUM	Premium Support Queue	"Thank you for calling. Please hold while we connect you."	< 120s
6d STANDARD	FLR Support Queue	Standard Exotel Care IVR prompt	Per existing SLA
6e UNKNOWN	Classification/Helpdesk Queue	"Please hold while we transfer your call."	Per existing SLA

Node 7 — Transfer to Queue with Call Context

When agent is selected and ringing, pass these variables as **call context payload** to the agent's CTI softphone:

```
{
  "account_name": "{{account_name}}",
  "support_tier": "{{support_tier}}",
  "rag_status": "{{rag_status}}",
  "csm_name": "{{csm_name}}",
  "caller_number": "{{caller_number}}",
  "lookup_source": "{{sf_response.lookup_source}}"
}
```

This payload is consumed by the CTI softphone (Open CTI adapter in Salesforce). See [04 — Screen Pop & Layouts](#) for consumption details.

PS confirmation: Exact field list that can be passed as call context varies by ECC version and CTI adapter version. PS to confirm.

5. Edge Cases & Fallbacks

Scenario	Handling
CLI private / anonymous	Skip lookup → UNKNOWN queue + flag for post-call review
CLI is an unknown number (not in SF)	UNKNOWN queue with a short prompt asking caller to provide account ID via DTMF
Middleware timeout (>2s)	STANDARD queue with no disruption to caller
Middleware returns tier = HYPERCARE but no agents logged in	ECC queue overflow rules take over — define overflow to Strategic queue → then Standard queue
Caller calls repeatedly (same ANI, back-to-back)	Cache ensures consistent tier; log repeated callers for follow-up
Number matches multiple accounts (SF query returned first)	Middleware logs warning, IVR proceeds with first match; Data team cleans up

International number (no +91)	Middleware attempts international lookup; if no match, STANDARD queue
-------------------------------	---

6. Testing Approach

6.1 — Pre-Launch Tests

Run each test by placing a real call from a test phone to the IVR DID.

#	Test	Expected Behavior
T1	Call from a registered Strategic customer phone	Routed to Strategic Support Queue, prompt played, agent sees context payload
T2	Call from Hypercare customer (flag active + future end date)	Routed to Hypercare Queue
T3	Call from Hypercare customer (flag active but end date expired)	Routed per base tier (NOT Hypercare)
T4	Call from Premium (SAAS Enterprise + RAG GREEN)	Premium Queue
T5	Call from unknown number	UNKNOWN / Helpdesk Queue
T6	Call from a number that exists in SF but tier = STANDARD	Standard FLR Queue
T7	Middleware disabled / down during test	STANDARD Queue, caller experience unaffected
T8	Middleware returns 500	STANDARD Queue
T9	Middleware slow (inject 3s delay)	Timeout after 2s → STANDARD Queue
T10	Private / masked CLI	UNKNOWN Queue

6.2 — Load Test

Middleware under load:

- Target: handle 100 req/sec sustained (BHCA max is 100k/hr = 28 req/sec across all customers)
- Cache hit ratio target: > 80% after warm-up
- p99 latency target: < 300ms

7. Monitoring & Observability

IVR-side

- Log every HTTP callout: request phone, response tier, response time, cache hit/miss
- Dashboard metric: % calls with successful tier lookup (target > 90%)
- Dashboard metric: % calls falling to STANDARD due to middleware failure (target < 2%)

Middleware-side

- Emit structured logs per request: phone, tier, source (cache/sf), latency, status
- Metrics: request rate, cache hit rate, SF API latency, error rate
- Alerting: SF API error rate > 5% over 5 min → page on-call

Salesforce-side

- Monitor API usage from Integration User; alert if nearing daily API limit
- Monthly report on tier distribution (see [06](#))

8. Security & Compliance

Concern	Mitigation
Salesforce credentials in IVR config	Option B middleware keeps creds out of IVR — stored in vault
Static shared secret between IVR and middleware	Rotate quarterly; restrict by source IP
Phone numbers logged in middleware	Consider hashing for audit logs; retention policy 30 days
Integration user permissions	Read-only on Account, Contact only; no write permissions
PII exposure	Only Account Name, CSM name, tier, RAG transmitted — no customer PII beyond phone

9. Rollback Plan

Rollback must be executable in < 5 minutes:

1. **Disable HTTP callout node** in IVR nodeflow → all calls fall through to existing queue routing
2. Or: **Flip feature flag** in middleware to always return `STANDARD` tier
3. Or: **Revert IVR nodeflow** to previous published version (both 4x and 6x support nodeflow versioning)

No Salesforce-side rollback needed — Case Flow operates independently and has its own rollback (deactivate Flow).

10. Open Questions for PS

See [05 — PS Engagement Brief](#) for the full list. Highlights:

1. Does the 4x / 6x HTTP action node support 2-second timeout with graceful fallback?
2. What is the max payload size for call context variables passed to agent?
3. Is there a native OAuth 2.0 flow in IVR nodeflow, or must we use static bearer token via middleware?
4. Can we pass structured JSON as call context, or must we pass key-value pairs?
5. What's the recommended pattern for queue overflow when tier-specific queue has no available agents?